INSTRUCTOR COPY · IN-PERSON EDITION · 45 MINUTES

FACILITATOR GUIDE

Room setup, the evidence link, timing, troubleshooting, and notes for every slide. [DEMO] marks a live demo in Excel.

THE SHAPE OF THE SESSION

Five timed sprints. Each one: a 1–1.5 minute briefing, a 3–5 minute work period with the timer on screen, and a 30-second checkpoint where you reveal the expected numbers. When the timer ends, **the room moves on together**. Anyone behind switches to the matching `Catch_Up` sheet. Everything not done in the room is documented in the *On Your Own* guide, with expected numbers so students can check themselves.

Time	Segment	Checkpoint to reveal
0:00–4:00	Welcome, how today works, the workbook	
4:00–9:00	Pre-flight: open link, make a copy, duplicate tab	Everyone has `Working`
9:00–14:00	Sprint 1 · Survey the scene	Clues from the room
14:00–21:00	Sprint 2 · Clear the scene	22 duplicates · 1,317 rows
21:00–28:00	Sprint 3 · Hidden characters	63 titles changed
28:00–35:00	Sprint 4 · Unmask the aliases	0 CHECK · 15 broadcasters · WDR 238
35:00–42:00	Sprint 5 · Time of the crime	5 impossible dates · first real date 29 Nov 1970
42:00–45:00	What's left, habits, exit ticket	

If you're running late: skip the Sprint 1 checkpoint reveal (say the clues yourself in 20 seconds) and shorten Sprint 5 to the `Date_Check` count. Never cut the pre-flight: everything depends on it.

BEFORE THE SESSION

- ☐ **Upload the evidence** `Tatort_Evidence.xlsx` to your OneDrive (office.com → Upload).
- ☐ **Make a view-only link:** in OneDrive, select the file → **Share** → link settings (the gear or "…") → *People in American University* (or *Anyone*) → **Can view** → Apply → **Copy link**. View-only is important: students can't accidentally edit your master, and they have to make their own copy.
- ☐ **Make a short link** that's easy to type and read aloud (e.g. a bit.ly or AU short link ending in `/tatort-excel`). Students type it into their **laptop** browser, so keep it short and unambiguous: avoid characters like `l` / `1` / `I` and `O` / `0`.
- ☐ **Put the short link everywhere:** on the Pre-flight slide (edit the placeholder in `source/slides.qmd` and re-render), in the blank on the Evidence Card before printing, on the whiteboard, and in advance via Canvas, the registration email or the calendar invite so people can simply click it.
- ☐ **Dry run** from a student-type account: open the link → **File → Create a Copy → Create a copy online** → check the copy opens in editing mode with all eight tabs. (The menu path was verified in editing mode on an AU account. Confirm it from a view-only link before the first session.)
- ☐ **Test the formula copy routine** once in the browser you'll present from (Formulas sheet → formula bar → Ctrl/⌘ + A, C → Esc → target cell → Ctrl/⌘ + V). If pasting gives text instead of a formula, tell students to click into the formula bar of the target cell before pasting.
- ☐ **Print** one Evidence Card per student (1 page) and a few cheat sheets. The On Your Own guide is for afterwards: share the link, no need to print.
- ☐ **Open** `Slides.html` in a browser, full screen. Press **S** for speaker view.

THE SPRINT TIMER

On every “Your Turn” slide a countdown starts automatically in the top-right corner. It turns red in the last minute and shows **TIME · CHECKPOINT** at zero. **Click it** to pause or resume. Press **T** to restart it. In the PDF the timer is replaced by a static label (e.g. *⏱ 5 min*).

TROUBLESHOOTING

Symptom	Likely cause and fix
“View only”, can’t type	They’re in your shared master. File → Create a Copy → Create a copy online.
Copy option missing	Fallback: File → Download a copy → office.com → Upload. Costs 2 minutes.
Remove Duplicates finds 0	Junk columns (`Import_ID`) still there. Delete them, run it again.
`#NAME?`	Header doesn’t match the formula: `TITEL` , `Folge` , `Sender` with a space. Or a typo in a function name.
Formula doesn’t fill down	They’re outside the Table. Check the new column is directly right of the table. Otherwise click the lightning icon → *Overwrite all cells*, or double-click the fill handle.
Formula pasted as text	They copied the cell, not the formula bar text. Redo from the formula bar, or type `=` and paste the rest.
“Browser can’t access the clipboard”	The Paste ▾ menu. Use the keyboard shortcuts instead.
Lots of CHECK in `Sender_Code`	Formula points at the wrong sheet or `TRIM` was left out. Recopy it.
Dates show as numbers or `#####`	Home → Number Format → Short Date; widen the column.
Counters give 0	They’re working in a `Catch_Up` sheet: replace `Working` in the counter with that sheet’s name.
Wrecked `Working`	Jump to the latest `Catch_Up` sheet. Their `Tatort_Evidence` is untouched, and File → Version History can roll back the whole file.

SLIDE-BY-SLIDE NOTES

2 SONNTAG, 20:15 UHR

0:00–2:00 · Welcome + hook

- While people sit down, the link to the evidence file should already be on screen or in the chat (see the Facilitator Guide). Ask everyone to sign in to office.com now.
- Quick show of hands: anyone watched Tatort? Real, big, multilingual data assembled by many hands over 55 years. That’s exactly how messy data happens.
- Fun fact from the dataset: episode 25, *Tote Taube in der Beethovenstraße* (1973), was directed by Samuel Fuller and ran in US cinemas.
- Hand out the Evidence Card now, one per person.

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3 HOW TODAY WORKS

2:00–3:00 · How today works

- This slide sets the routine for the session. Short briefings, then they work on their own while the timer runs. When the timer hits zero, we all compare numbers and move on together, wherever you are.
- Point out the catch-up sheets: nobody falls more than one sprint behind. Make it clear that switching to a catch-up sheet is expected.
- Some of the case is left for later. The “On Your Own” guide documents every remaining step.

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4 MEET THE EVIDENCE

3:00–4:00 · Tour of the workbook

- Point out the Formulas sheet and how to copy from it: click the green cell, click in the formula bar, Ctrl/⌘+A then Ctrl/⌘+C, Esc. Click the target cell, Ctrl/⌘+V. Demo it once in Sprint 3; don't spend time on it now.
- The grey tabs at the end are the catch-up sheets. Don't open them yet: they're spoilers for the next sprint.
- The data is real, but errors were planted deliberately. Don't cite it as a source of Tatort facts!

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5 PRE-FLIGHT

4:00–9:00 · Pre-flight (4-min timer + 1 min slack)

- This is where in-person workshops lose 10 minutes. Show the short link on screen, read it aloud, and write it on the whiteboard. If you sent it in advance (Canvas, email), people can click it there instead.
- The shared file is view-only. Everyone must make their own copy; if anyone sees "Editing" on the shared file itself, stop them (the Facilitator Guide explains how to set the link to view-only so this can't happen).
- If the copy option is missing for someone, fallback: File → Download, then upload to their own OneDrive (office.com → Upload).
- Duplicate tab = chain of custody. In the browser there's no "close without saving".
- Walk the room. When most hands are up, move on. Anyone still stuck: pair them with a neighbour for now and come back to them.

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6 SPRINT 1 · LOOK BEFORE YOU TOUCH

9:00–10:00 · Briefing (1 min)

- [DEMO, 30 s] Data → Filter, open the `Sender` drop-down: `wdr`, `W.D.R.`, `Westdeutscher Rundfunk` … This usually convinces people that filters are worth using.
- Don't fix anything in this sprint. Just look.

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7 SPRINT 1: SURVEY THE SCENE

10:00–13:00 · Work (3 min)

- Walk the room. Look for people who are lost in the ribbon: point them to the search box.
- Early finishers: find a fourth and fifth clue, or compare `LEN` of a few titles.

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8 CHECKPOINT 1

13:00–14:00 · Checkpoint (1 min)

- Take 3–4 clues from the room, then reveal the list one line at a time (each click shows the next).
- Name the plan: sprint 2 clears the scene, 3 hidden characters, 4 aliases, 5 dates. Everything else is documented for later.

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9 SPRINT 2 · CLEAR THE SCENE

14:00–15:30 · Briefing (1.5 min)

- Why title rows matter: they break filters, tables and pivot tables.
- Why a Table: from Sprint 3 on, every formula fills down the whole column by itself, which saves a lot of time in a 45-minute session.
- Remove Duplicates with all columns = only rows identical in every cell. Choosing fewer columns is a judgement call about what "the same record" means.

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10 SPRINT 2: CLEAR THE SCENE

15:30–20:30 · Work (5 min)

- Common trap: deleting the junk columns *after* Remove Duplicates. `Import_ID` is different on every row, so Excel finds 0 duplicates. Order matters: clear the scene first.
- Headers matter now because the formulas in the next sprints use them by name (`[@Titel]`). A header with a stray space breaks the formula.
- Ignore the yellow highlights for now (On Your Own guide).

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11 CHECKPOINT 2

20:30–21:00 · Checkpoint (30 s)

- Hands up for 22. Anyone with a different number: switch to Catch_Up_2 now and carry on. Fix the mistake later, not now.
- Catch_Up sheets are already Tables, so new columns fill down there too.

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12 SPRINT 3 · HIDDEN CHARACTERS

21:00–22:30 · Briefing + the one demo that matters (1.5 min)

- [DEMO] The copy routine from the Formulas sheet, once, slowly: click the green cell → click in the formula bar → Ctrl/⌘+A, Ctrl/⌘+C → Esc → type `Titel_Clean` in the first empty header cell to the right of the table → Enter → click the cell below → Ctrl/⌘+V → Enter. The whole column fills.
- If the column doesn't fill by itself: a small lightning icon offers "Overwrite all cells in this column". Or double-click the fill handle.
- We keep the helper columns rather than pasting values back. Fewer steps, and the original stays beside the clean version for comparison. The On Your Own guide shows how to swap them in at the end.

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13 SPRINT 3: HIDDEN CHARACTERS

22:30–27:30 · Work (5 min)

- Seeing `#NAME?` usually means a header doesn't match: `Titel` vs `TITEL` , or a stray space. Check the header row.
- If someone typed `CHAR(160)` instead of `UNICHAR(160)` and sees fewer changes, that's the first thing to check.
- Fast finishers: filter `Titel_Clean` for a title that changed and compare it with the original in the formula bar.

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14 CHECKPOINT 3

27:30–28:00 · Checkpoint (30 s)

- 63 is the number. 57 or 51 means the non-breaking spaces or the double spaces weren't caught.
- Clear the TRUE filter before the next sprint (Data → Clear, or the filter drop-down → Clear Filter).

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15 SPRINT 4 · UNMASK THE ALIASES

28:00–29:30 · Briefing (1.5 min)

- Read XLOOKUP aloud in plain English: "look up this broadcaster in column A of Known_Aliases and bring back its code from column B; if it isn't there, write CHECK."
- XLOOKUP ignores capitalisation, so `wdr` matches `WDR` . That's why the alias list doesn't need every case variant.
- The empty arguments (`, , , ,`) skip options we don't need. The last one says "if there's no slash, use the whole cell". They'll copy it, not type it.
- Reference tables are useful well beyond this dataset: majors, country names, department codes, survey answers…

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16 SPRINT 4: UNMASK THE ALIASES

29:30–34:30 · Work (5 min)

- The counters use column letters (K, L, N), which is why everyone adds the columns in the same order. In a Catch_Up sheet, replace `Working` with the sheet name.
- Anyone with lots of CHECK: they probably skipped the `TRIM` (trailing spaces) or the formula points at the wrong sheet.
- Fast finishers: Insert → PivotTable → `Sender_Code` in Rows and in Values. A whole broadcaster count in 20 seconds (On Your Own, Lead H).

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17 CHECKPOINT 4

34:30–35:00 · Checkpoint (30 s)

- 238 for WDR. (None of the 6 remaining near-duplicates is a WDR episode, so this number is already final.)
- Clear the filter before the last sprint.

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18 SPRINT 5 · TIME OF THE CRIME

35:00–36:30 · Briefing (1.5 min)

- Excel stores dates as numbers: days since 1900. So `25901` is 29 November 1970, the very first Tatort.
- DATEVALUE guesses using the computer's regional settings; DATE never guesses.
- The formula on the Formulas sheet uses LET: `d` is a nickname for the date cell so it doesn't have to be repeated. Read it aloud: already a date? keep it. Third character a dot? German. Otherwise ISO.
- Plausibility checks: you can't always know the right value, but you can tell when a value is impossible. Flag it rather than guessing a fix.

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19 SPRINT 5: TIME OF THE CRIME

36:30–41:30 · Work (5 min)

- A date showing as `#####` just means the column is too narrow: double-click the column border.
- One impossible date comes out as the year 3799: the typo `1899` plus Excel's rule that years below 1900 get 1900 added. A great "don't trust the output" moment if someone spots it.
- Don't fix the impossible dates. Flag them in the Case Log; the right date has to come from a source.

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20 CHECKPOINT 5

41:30–42:00 · Checkpoint (30 s)

- If the first date is in 1907 or 1958, they sorted before filtering out the CHECK rows. Sorting puts the impossible dates at the top, which is why it's a useful check.

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21 CASE NOT CLOSED (YET)

42:00–43:00 · What's left

- Each lead takes 3–5 minutes with the guide; they can do it later today while it's fresh. The guide has the formulas and the expected numbers, so they can check themselves.
- Their copy stays in their OneDrive, so they can pick up exactly where they left off.

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22 GOOD HABITS

43:00–44:00 · Wrap-up

- These habits carry straight over to R, Python or anything else. Mention the R + AI workshop for anyone who wants to automate this.

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23 BEFORE YOU GO

44:00–45:00 · Exit ticket

- Collect the cards at the door. They tell you what stuck and what to fix next time.
- Remind people the guide and cheat sheet are in the repo, CC BY 4.0.
